

- Some patients may experience apparent lack of consistent efficacy due to activation of latent or underlying bipolar disorder, and require antidepressant discontinuation and switch to a mood stabilizer

Best Augmenting Combos for Partial Response or Treatment Resistance

- Augmentation experience is limited compared to other antidepressants
- Benzodiazepines can reduce insomnia and anxiety
- Bupropion, mirtazapine, reboxetine, or atomoxetine (use combinations of antidepressants with caution as this may activate bipolar disorder and suicidal ideation)
- Modafinil, especially for fatigue, sleepiness, and lack of concentration
- Mood stabilizers or atypical antipsychotics for bipolar depression, psychotic depression, or treatment-resistant depression
- Hypnotics or trazodone for insomnia
- Classically, lithium, buspirone, or thyroid hormone

Tests

- Check heart rate and blood pressure before initiating treatment and regularly during treatment

Side Effects

How Drug Causes Side Effects

- Theoretically due to increases in serotonin and norepinephrine concentrations at receptors in parts of the brain and body other than those that cause therapeutic actions (e.g., unwanted actions of serotonin in sleep centers causing insomnia, unwanted actions of norepinephrine on acetylcholine release causing urinary retention or constipation)
- Most side effects are immediate but often go away with time, in contrast to most therapeutic effects, which are delayed and are enhanced over time

Notable Side Effects

- Nausea, vomiting, constipation
- Hyperhidrosis
- Tachycardia, heart rate increase, palpitations
- Erectile dysfunction
- Urinary hesitancy or retention



Life-Threatening or Dangerous Side Effects

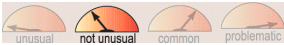
- Rare seizures
- Rare induction of mania and activation of suicidal ideation

Weight Gain



- Reported but not expected

Sedation



- Occurs in significant minority

What to Do About Side Effects

- Wait
- Wait
- Wait
- Lower the dose
- In a few weeks, switch to another agent or add other drugs

Best Augmenting Agents for Side Effects

* For urinary hesitancy, give an alpha 1 blocker such as tamsulosin or naftopidil

- Often best to try another antidepressant monotherapy prior to resorting to augmentation strategies to treat side effects
- Trazodone or a hypnotic for insomnia
- Bupropion, sildenafil, vardenafil, or tadalafil for sexual dysfunction
- Benzodiazepines for anxiety, agitation

- Mirtazapine for insomnia, agitation, and gastrointestinal side effects
- Many side effects are time-dependent (i.e., they start immediately upon dosing and upon each dose increase, but go away with time)
- Activation and agitation may represent the induction of a bipolar state, especially a mixed dysphoric bipolar II condition sometimes associated with suicidal ideation, and require the addition of lithium, a mood stabilizer or an atypical antipsychotic, and/or discontinuation of levomilnacipran

Dosing and Use

Usual Dosage Range

- 40–120 mg once daily

Dosage Forms

- Extended-release capsule 20 mg, 40 mg, 80 mg, 120 mg

How to Dose

- Initial dose 20 mg once daily for 2 days, then increase to 40 mg once daily; can increase by 40 mg/day every 2 or more days; maximum recommended dose 120 mg once daily



Dosing Tips

- Can be taken with or without food

- Do not break or chew levomilnacipran capsules, as this will alter controlled-release properties
- If intolerable anxiety, insomnia, agitation, akathisia, or activation occur upon either dosing initiation or discontinuation, consider the possibility of activating a bipolar disorder and switch to a mood stabilizer or an atypical antipsychotic

Overdose

- No fatalities have been reported; experience is limited

Long-Term Use

- Has not been evaluated in controlled studies, but long-term treatment of major depressive disorder is generally necessary

Habit Forming

- No

How to Stop

- Taper is prudent, but usually not necessary

Pharmacokinetics

- Metabolized by CYP450 3A4; renally excreted
- Terminal elimination half-life approximately 12 hours

Drug Interactions

- Tramadol increases the risk of seizures in patients taking an antidepressant
- Can cause a fatal “serotonin syndrome” when combined with MAOIs, so do not use with MAOIs or for at least 14 days after MAOIs are stopped
- Do not start an MAOI for at least 5 half-lives (5 to 7 days for most drugs) after discontinuing levomilnacipran
- Strong CYP450 3A4 inhibitors, such as ketoconazole, can increase plasma levels of levomilnacipran; do not exceed 80 mg once daily of levomilnacipran if used with a strong CYP450 3A4 inhibitor
- Alcohol may interact with the extended-release properties of levomilnacipran, causing a pronounced accelerated drug release (“drug dumping”); thus, taking levomilnacipran with alcohol is not recommended
- Possible increased risk of bleeding, especially when combined with anticoagulants (e.g., warfarin, NSAIDs)
- Switching from or addition of other norepinephrine reuptake inhibitors should be done with caution, as the additive pro-noradrenergic effects may enhance therapeutic actions in depression, but also enhance noradrenergically mediated side effects

Other Warnings/Precautions

- Use with caution in patients with history of seizures

- Use with caution in patients with bipolar disorder unless treated with concomitant mood-stabilizing agent
- Use with caution in patients with controlled angle-closure glaucoma
- Not approved in children, so when treating children off label, carefully weigh the risks and benefits of pharmacological treatment against the risks and benefits of non-treatment with antidepressants and make sure to document this in the patient's chart
- Distribute the brochures provided by the FDA and the drug companies
- Whenever possible, warn patients and their caregivers about the possibility of activating side effects, and advise them to report such symptoms immediately
- Monitor patients for activation of suicidal ideation, especially children and adolescents

Do Not Use

- If patient has uncontrolled angle-closure glaucoma
- If patient is taking an MAOI
- If there is a proven allergy to levomilnacipran or milnacipran

Special Populations

Renal Impairment

- Maximum dose 80 mg once daily for moderate impairment
- Maximum dose 40 mg once daily for severe impairment

Hepatic Impairment

- Dose adjustment not necessary

Cardiac Impairment

- Not systematically evaluated in patients with cardiac impairment
- Drug should be used with caution

Elderly

- Some patients may tolerate lower doses better
- Reduction in risk of suicidality with antidepressants compared to placebo in adults age 65 and older

Children and Adolescents

- Safety and efficacy have not been established
- Carefully weigh the risks and benefits of pharmacological treatment against the risks and benefits of nontreatment with antidepressants and make sure to document this in the patient's chart
- Monitor patients face-to-face regularly, particularly during the first several weeks of treatment

- Use with caution, observing for activation of known or unknown bipolar disorder and/or suicidal ideation, and strongly consider informing parents or guardian of this risk so they can help observe child or adolescent patients

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- Not generally recommended for use during pregnancy, especially during first trimester
- Nonetheless, continuous treatment during pregnancy may be necessary and has not been proven to be harmful to the fetus
- At delivery there may be more bleeding in the mother and transient irritability or sedation in the newborn
- Must weigh the risk of treatment (first trimester fetal development, third trimester newborn delivery) to the child against the risk of no treatment (recurrence of depression, maternal health, infant bonding) to the mother and child

- For many patients, this may mean continuing treatment during pregnancy
- Exposure to serotonin reuptake inhibitors early in pregnancy may be associated with increased risk of septal heart defects (absolute risk is small)
- Use of serotonin reuptake inhibitors beyond the 20th week of pregnancy may be associated with increased risk of pulmonary hypertension in newborns, although this is not proven
- Exposure to serotonin reuptake inhibitors late in pregnancy may be associated with increased risk of gestational hypertension and preeclampsia
- Neonates exposed to SSRIs or SNRIs late in the third trimester have developed complications requiring prolonged hospitalization, respiratory support, and tube feeding; reported symptoms are consistent with either a direct toxic effect of SSRIs and SNRIs or, possibly, a drug discontinuation syndrome, and include respiratory distress, cyanosis, apnea, seizures, temperature instability, feeding difficulty, vomiting, hypoglycemia, hypotonia, hypertonia, hyperreflexia, tremor, jitteriness, irritability, and constant crying
- National Pregnancy Registry for Antidepressants: 1-866-961-2388, <https://womensmentalhealth.org/research/pregnancyregistry/antidepressants>

Breast Feeding

- Unknown if levomilnacipran is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- If child becomes irritable or sedated, breast feeding or drug may need to be discontinued
- Immediate postpartum period is a high-risk time for depression, especially in women who have had prior depressive episodes, so drug may need to be reinstituted late in the third trimester or shortly after childbirth to prevent a recurrence during the postpartum period
- Must weigh benefits of breast feeding with risks and benefits of antidepressant treatment versus nontreatment to both the infant and the mother
- For many patients, this may mean continuing treatment during breast feeding

The Art of Psychopharmacology

Potential Advantages

- Patients with depression may have higher remission rates on SNRIs than on SSRIs
- Depressed patients with somatic symptoms, fatigue, and pain

Potential Disadvantages

- Cost
- Patients with urologic disorders, prostate disorders

- Patients with borderline or uncontrolled hypertension
- Patients with agitation and anxiety (short-term)

Primary Target Symptoms

- Depressed mood
- Physical symptoms

Pearls

- * Has greater potency for norepinephrine reuptake blockade than for serotonin reuptake blockade, but this is of unclear clinical significance as a differentiating feature from other SNRIs, although it might contribute to theoretical effects in fibromyalgia and chronic pain
- * Potent noradrenergic actions may account for possibly higher incidence of sweating and urinary hesitancy than some other SNRIs
- Urinary hesitancy more common in men than women and in older men than in younger men
- Alpha 1 antagonists such as tamsulosin or naftopidil can reverse urinary hesitancy or retention
- Alpha 1 antagonists given prophylactically may prevent urinary hesitancy or retention in patients at higher risk, such as elderly men with borderline urine flow
- Nonresponse to levomilnacipran in elderly may require consideration of mild cognitive impairment or Alzheimer disease

Suggested Reading

Auclair AL , Martel JC , AssiÃ© MB , et al. Levomilnacipran (F2695), a norepinephrine-preferring SNRI: profile in vitro and in models of depression and anxiety . *Neuropharmacology* 2013 ;**70** :338–47 .

Citrome L. Levomilnacipran for major depressive disorder: a systematic review of the efficacy and safety profile for this newly approved antidepressant –what is the number needed to treat, number needed to harm and likelihood to be helped or harmed? *Int J Clin Pract* 2013 ;**67** :1089 – 104.

Mago R , Forero G , Greenberg WM , Gommoll C , Chen C. Safety and tolerability of levomilnacipran ER in major depressive disorder: results from an open-label, 48-week extension study . *Clin Drug Investig* 2013 ;**33** :761–71 .

Lisdexamfetamine

Therapeutics

Brands

- Vyvanse
- *see index for additional brand names*

Generic?

No

Class

- Neuroscience-based Nomenclature: dopamine and norepinephrine reuptake inhibitor and releaser (DN-RIRe)
- Stimulant

Commonly Prescribed for

(bold for FDA approved)

- **Attention deficit hyperactivity disorder (ADHD) (ages 6 and older)**
- **Binge eating disorder**
- Treatment-resistant depression

How the Drug Works

- * Lisdexamfetamine is a prodrug of dextroamphetamine and is thus not active until after it has been absorbed by the intestinal tract and converted to dextroamphetamine (active component) and l-lysine
- * Once converted to dextroamphetamine, it increases norepinephrine and especially dopamine actions by blocking their reuptake and facilitating their release
- Enhancement of dopamine and norepinephrine in certain brain regions (e.g., dorsolateral prefrontal cortex) may improve attention, concentration, executive dysfunction, and wakefulness
- Enhancement of dopamine actions in other brain regions (e.g., basal ganglia) may improve hyperactivity
- Enhancement of dopamine and norepinephrine in yet other brain regions (e.g., medial prefrontal cortex, hypothalamus) may improve depression, fatigue, and sleepiness

How Long Until It Works

- Some immediate effects can be seen with first dosing
- Can take several weeks to attain maximum therapeutic benefit

If It Works (for ADHD)

- The goal of treatment of ADHD is reduction of symptoms of inattentiveness, motor hyperactivity, and/or impulsiveness that disrupt social, school, and/or occupational functioning

- Continue treatment until all symptoms are under control or improvement is stable and then continue treatment indefinitely as long as improvement persists
- Reevaluate the need for treatment periodically
- Treatment for ADHD begun in childhood may need to be continued into adolescence and adulthood if continued benefit is documented

If It Doesn't Work (for ADHD)

- Consider adjusting dose or switching to another formulation of d-amphetamine or to another agent
 - Consider behavioral therapy
 - Consider the presence of noncompliance and counsel patients and parents
 - Consider evaluation for another diagnosis or for a comorbid condition (e.g., bipolar disorder, substance abuse, medical illness, etc.)
- * Some ADHD patients and some depressed patients may experience lack of consistent efficacy due to activation of latent or underlying bipolar disorder, and require either augmenting with a mood stabilizer or switching to a mood stabilizer



Best Augmenting Combos for Partial Response or Treatment Resistance

- * Best to attempt other monotherapies prior to augmenting

- For the expert, can combine with modafinil or atomoxetine for ADHD
- For the expert, can occasionally combine with atypical antipsychotics in highly treatment-resistant cases of bipolar disorder or ADHD
- For the expert, can combine with antidepressants to boost antidepressant efficacy in highly treatment-resistant cases of depression while carefully monitoring patient

Tests

- Before treatment, assess for presence of cardiac disease (history, family history, physical exam)
- Blood pressure should be monitored regularly
- In children, monitor weight and height

Side Effects

How Drug Causes Side Effects

- Increases in norepinephrine especially peripherally can cause autonomic side effects, including tremor, tachycardia, hypertension, and cardiac arrhythmias
- Increases in norepinephrine and dopamine centrally can cause CNS side effects such as insomnia, agitation, psychosis, and substance abuse

Notable Side Effects

- * Insomnia, headache, exacerbation of tics, nervousness, irritability, overstimulation, tremor, dizziness
- * Anorexia, nausea, dry mouth, constipation, diarrhea, weight loss
- Can temporarily slow normal growth in children (controversial)
- Sexual dysfunction long-term (impotence, libido changes), but can also improve sexual dysfunction short-term



Life-Threatening or Dangerous Side Effects

- Psychotic episodes
- Seizures
- Palpitations, tachycardia, hypertension
- Rare activation of hypomania, mania, or suicidal ideation (controversial)
- Cardiovascular adverse effects, sudden death in patients with preexisting cardiac structural abnormalities

Weight Gain



- Reported but not expected
- Some patients may experience weight loss

Sedation



- Reported but not expected
- Activation much more common than sedation

What to Do About Side Effects

- Wait
- Adjust dose
- Switch to another long-acting stimulant
- Switch to another agent
- For insomnia, avoid dosing in afternoon/evening

Best Augmenting Agents for Side Effects

- Beta blockers for peripheral autonomic side effects
- Dose reduction or switching to another agent may be more effective since most side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- ADHD: 30–70 mg/day
- Binge eating disorder: 50–70 mg/day

Dosage Forms

- Capsule 10 mg, 20 mg, 30 mg, 40 mg, 50 mg, 60 mg, 70 mg
- Chewable tablet 10 mg, 20 mg, 30 mg, 40 mg, 50 mg, 60 mg

How to Dose

- Initial 30 mg/day in the morning; can increase by 10–20 mg each week; maximum dose generally 70 mg/day
- Binge eating disorder: initial 30 mg/day in the morning; can increase by 20 mg each week; maximum dose generally 70 mg/day



Dosing Tips

- 10–12 hour duration of clinical action
- Capsules can either be taken whole or they can be opened and the contents dissolved in water
- When taken in water, the entire solution should be consumed immediately
- Dose of a single capsule should not be divided
- Once daily dosing can be an important practical element in stimulant utilization, eliminating the hassle and pragmatic difficulties of lunchtime dosing at school, including storage problems, potential diversion, and the need for a medical professional to supervise dosing away from home
- Avoid dosing after the morning because of the risk of insomnia

- * May be possible to dose only during the school week for some ADHD patients
- * May be able to give drug holidays for patients with ADHD over the summer in order to reassess therapeutic utility and effects on growth suppression as well as to assess any other side effects and the need to reinstitute stimulant treatment for the next school term
- However, most studies show that parental height is what determines a patient's final height, and that most children/adolescents taking stimulants reach their expected height, just more slowly than children/adolescents not exposed to stimulants
- Can be taken with or without food

Overdose

- Rarely fatal; panic, hyperreflexia, rhabdomyolysis, rapid respiration, confusion, coma, hallucination, convulsion, arrhythmia, change in blood pressure, circulatory collapse

Long-Term Use

- Can be used long-term for ADHD when ongoing monitoring documents continued efficacy
- Dependence and/or abuse may develop
- Tolerance to therapeutic effects may develop in some patients
- Long-term stimulant use may be associated with growth suppression in children (controversial)

- Periodic monitoring of weight, blood pressure, CBC, platelet counts, and liver function may be prudent

Habit Forming

- Schedule II drug
- Patients may develop tolerance, psychological dependence
- Theoretically less abuse potential than other stimulants when taken as directed because it is inactive until it reaches the gut and thus has delayed time to onset as well as long duration of action

How to Stop

- Taper to avoid withdrawal effects
- Withdrawal following chronic therapeutic use may unmask symptoms of the underlying disorder and may require follow-up and reinstitution of treatment
- Careful supervision is required during withdrawal from abuse use since severe depression may occur

Pharmacokinetics

- 1 hour to maximum concentration of lisdexamfetamine, 3.5 hours to maximum concentration of dextroamphetamine
- Duration of clinical action 10–12 hours



Drug Interactions

- May affect blood pressure and should be used cautiously with agents used to control blood pressure
- Gastrointestinal acidifying agents (guanethidine, reserpine, glutamic acid, ascorbic acid, fruit juices, etc.) and urinary acidifying agents (ammonium chloride, sodium phosphate, etc.) lower amphetamine plasma levels, so such agents can be useful to administer after an overdose but may also lower therapeutic efficacy of amphetamines
- Gastrointestinal alkalinizing agents (sodium bicarbonate, etc.) and urinary alkalinizing agents (acetazolamide, some thiazides) increase amphetamine plasma levels and potentiate amphetamine's actions
- Desipramine and protryptiline can cause striking and sustained increases in brain concentrations of d-amphetamine and may also add to d-amphetamine's cardiovascular effects
- Theoretically, other agents with norepinephrine reuptake blocking properties, such as venlafaxine, duloxetine, atomoxetine, milnacipran, and reboxetine, could also add to amphetamine's CNS and cardiovascular effects
- Amphetamines may counteract the sedative effects of antihistamines
- Haloperidol, chlorpromazine, and lithium may inhibit stimulatory effects of amphetamine
- Theoretically, atypical antipsychotics should also inhibit stimulatory effects of amphetamines
- Theoretically, amphetamines could inhibit the antipsychotic actions of antipsychotics

- Theoretically, amphetamines could inhibit the mood-stabilizing actions of atypical antipsychotics in some patients
- Combinations of amphetamines with mood stabilizers (lithium, anticonvulsants, atypical antipsychotics) is generally something for experts only, when monitoring patients closely and when other options fail
- Absorption of phenobarbital, phenytoin, and ethosuximide is delayed by amphetamines
- Amphetamines inhibit adrenergic blockers and enhance adrenergic effects of norepinephrine
- Amphetamines may antagonize hypotensive effects of Veratrum alkaloids and other antihypertensives
- Amphetamines increase the analgesic effects of meperidine
- Amphetamines contribute to excessive CNS stimulation if used with large doses of propoxyphene
- Amphetamines can raise plasma corticosteroid levels
- MAOIs slow absorption of amphetamines and thus potentiate their actions, which can cause headache, hypertension, and rarely hypertensive crisis and malignant hyperthermia, sometimes with fatal results
- Use with MAOIs, including within 14 days of MAOI use, is not advised, but this can sometimes be considered by experts who monitor depressed patients closely when other treatment options for depression fail

Other Warnings/Precautions

- Use with caution in patients with any degree of hypertension, hyperthyroidism, or history of drug abuse
- Children who are not growing or gaining weight should stop treatment, at least temporarily
- May worsen motor and phonic tics
- May worsen symptoms of thought disorder and behavior disturbance in psychotic patients
- Stimulants have a high potential for abuse and must be used with caution in anyone with a current or past history of substance abuse or alcoholism or in emotionally unstable patients
- Administration of stimulants for prolonged periods of time should be avoided whenever possible or done only with close monitoring, as it may lead to marked tolerance and drug dependence, including psychological dependence with varying degrees of abnormal behavior
- Particular attention should be paid to the possibility of subjects obtaining stimulants for nontherapeutic use or distribution to others and the drugs should in general be prescribed sparingly with documentation of appropriate use
- Usual dosing has been associated with sudden death in children with structural cardiac abnormalities
- Not an appropriate first-line treatment for depression or for normal fatigue